

Adam Dejen

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EDUCATION

University of Bath

BEng Integrated Design (Mechanical) Engineering
Grade: 2:1

Sep. 2020 – July. 2025*

Bath, UK

* 2 years suspended academics due to health reasons

Gunnersbury Catholic School

GCSEs (8A*s 2As), A-levels (Maths, Further Maths & Physics: A*AA)

Sep. 2012 – Jun. 2019

London, UK

SKILLS + INTERESTS

I have a deep curiosity for how things work, which fuels my passion for hands-on engineering, sustainable design, and high-precision manufacturing. Whether it is prototyping, 3D modelling, or optimising systems, I enjoy solving tangible problems with creativity and technical insight.

Outside of engineering, I stay active through competitive football and athletics, which have taught me discipline and teamwork, and explore storytelling through screenwriting. I am always eager to learn, build, and contribute to projects that make a meaningful impact.

Extra-curricular: Screen-play writing, Competitive football Football, Competitive athletics (1500m)

Software: Microsoft Office, Autodesk Inventor, AutoCAD, Blender

Programming Languages: Python, MATLAB, C++, Excel, Inventor API

Languages: English (Native), Amharic (Proficient)

EXPERIENCE

UI Design Intern

Kleeder (Start-up)

May. 2018 – Aug. 2018

London, UK

- * Collaborated on UI/UX design for a service industry app using Figma and Adobe XD.
- * Contributed to marketing strategies through audience analysis and social media planning.

Key Stage 2/3 Maths Tutor

White City community center and Spaero school

Jan. 2018 – June. 2020

London, UK

- * Delivered tailored Math and English lessons, adapting to individual student needs.
- * Tracked academic progress and adjusted teaching strategies through regular assessments.

Specialist

Apple

Aug. 2024 – Jan. 2025

Bath, UK

- * Delivered expert product support to customers, enhancing the Apple ecosystem experience.
- * Collaborated with team to exceed store sales targets and maintain service excellence.

COMPETITIONS

E-Commerce Packaging Design Competition

University of Bath representative

UK

Dec. 2024

- * Designed and pitched an innovative packaging solution for an e-commerce clothing brand, focusing on formal-wear.
- * Utilised the Double Diamond methodology for structured idea development and refinement.
- * Led the final design phase using the "Six Thinking Hats" tool to ensure customer appeal and functionality.
- * Created a recyclable, low-waste packaging design that minimised garment creasing and enhanced the unboxing experience.

Development and Integration of a Gantry-Based Magnetic Scanning System

(1st grade) Unit: Mechatronic Design

Bath, UK

- * Designed, programmed, and integrated a gantry control system for precise scanning and handling of magnets using Hall Effect sensors and a gripper mechanism.
- * Developed S-pattern motion logic and homing sequence for gantry motors using Arduino and MATLAB.
- * Integrated Hall Effect sensors for real-time magnetic field detection and data visualisation.
- * Implemented a nearest-neighbor algorithm for magnet localisation, enhancing system precision.
- * Controlled an electromagnet and rack-and-pinion mechanism for object handling.
- * Conducted iterative testing, debugging, and performance optimisation to meet design requirements.

* **Skills Gained:**

- **Technical Skills:** Arduino programming, MATLAB scripting, motion control design, sensor integration, and system-level debugging.
- **Soft Skills:** Problem-solving, iterative development, team collaboration, effective documentation, and reporting.

Human Centred Design - Design and Prototyping of a Novel Extendable Bike Stem

(2:1 grade) Unit: Human Centred Design

Bath, UK

- * Led the prototyping design (high fidelity) and conducted market research for a user-centred design project focusing on cycling ergonomics.
- * Collaborated with the team to conduct user interviews, analyse requirements, and establish the project aim to address user needs effectively.
- * Presented mixed prototypes using a variety of methods, including CAD models, low and high-fidelity physical prototypes, and renders, showcasing design progress and working principles.
- * Iterated and refined the extendable bike stem design through iterative prototyping and testing, ensuring alignment with user requirements and functionality.
- * Articulated the design journey, challenges, and future steps in a detailed project report and presentation.

* **Skills Gained:**

- **Technical Skills:** CAD modeling, prototyping techniques, user-centred design methodology, and market research analysis.
- **Soft Skills:** Communication, teamwork, user interviewing, and project documentation.

Large Language Model (LLM)-Enabled Aerospace Mirror Design

(2:1 grade) Unit: Final Year Project

Bath, UK

- * Developed a framework to integrate GPT-4 with Autodesk Inventor's API using Python, automating parametric CAD generation for an aerospace mirror.
- * Scripted geometric operations (e.g., parabolic profile sketching) via Inventor API, resolving enumeration errors through manual debugging.
- * Implemented Chain-of-Thought prompting to improve GPT-4's spatial reasoning for CAD tasks.
- * Conducted FEA-driven optimisation (modal analysis) by iteratively refining designs based on GPT-4's structural suggestions (e.g., rib thickness adjustments).
- * Evaluated GPT-4's additive manufacturing recommendations (e.g., support structures, print orientation) against technician-executed 3D prints, identifying trade-offs between AI-proposed quality and practical material efficiency.

* **Skills Gained:**

- **Technical Skills:** Autodesk Inventor API, Python, GPT-4, Finite Element Analysis (FEA), ABS 3D Printing.
- **Soft Skills:** Project documentation, Prompting.